Example 2.2: Schmidt number and entanglement with qubits

Consider two normalised states of a bipartite system of two qubits

$$|\psi_1\rangle = \frac{1}{\sqrt{2}}|0\rangle_A|0\rangle_B + \frac{1}{\sqrt{2}}|0\rangle_A|1\rangle_B, \tag{2.59}$$

and

$$|\psi_2\rangle = \frac{1}{\sqrt{2}}|0\rangle_A|0\rangle_B + \frac{1}{\sqrt{2}}|1\rangle_A|1\rangle_B. \tag{2.60}$$

Are these states entangled?

Consider first $|\psi_1\rangle$. Recall that the $+1$ eigenstate of $\hat{\sigma}_x$ is $|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$ and note that $|\psi_1\rangle = |0\rangle_A|+\rangle_B$. This state is thus a product. Hence it is separable, i.e. *not* entangled. Furthermore, choosing the z-basis for qubit A and the x-basis for qubit B and denoting $|0'\rangle_B = |+\rangle_B$ we can then write the state on Schmidt form, $|\psi_1\rangle = |0\rangle_A|0'\rangle_B$. We see that the Schmidt number of $|\psi_1\rangle$ is 1.

Consider now $|\psi_2\rangle$. Because the coefficients in (2.60) are real and positive and the state is written using orthonormal bases for A and B, the state is already on Schmidt form. We thus see that the Schmidt number of $|\psi_2\rangle$ is 2 and this state *is* entangled.

2.3.3 Partial trace

Imagine that we have access only to a part of some larger system. What information is available through observations of this subsystem alone?

Let us denote our subsystem by A and the unaccessible part of the total system by B. We can model any accessible observable as taking the form

$$\hat{M}_A \otimes \mathbb{1}_B, \tag{2.61}$$

where $\hat{M}_A$ is any obervable on A and the identity can be understood as a completely uniformative measurement on B (it always outputs 1 independent of the state, and the corresponding eigenspace is all of the state space of B, so measuring $\mathbb{1}_B$ is equivalent to doing nothing). The expectation value of this observable for a general bipartite state (2.32) is

$$\begin{aligned}
\langle \hat{M}_A \rangle &= {}_{AB}\langle\psi|(\hat{M}_A \otimes \mathbb{1}_B)|\psi\rangle_{AB} && (2.62)\\
&= \sum_{n,n'}\sum_{m,m'} \alpha_{nm}\alpha^*_{n'm'}\,{}_{AB}\langle n', m'|(\hat{M}_A \otimes \mathbb{1}_B)|n, m\rangle_{AB} && (2.63)\\
&= \sum_{n,n'}\sum_{m} \alpha_{nm}\alpha^*_{n'm}\,{}_{AB}\langle n'|\hat{M}_A|n\rangle_{AB} && (2.64)\\
&= \mathrm{Tr}[\hat{M}_A\hat{\rho}_A], && (2.65)
\end{aligned}$$

where we have used that $\{|m\rangle_B\}_m$ is an orthonormal basis for B and we have defined the operator on subsystem A

$$\hat{\rho}_A = \sum_{n,n'}\sum_{m} \alpha_{nm}\alpha^*_{n'm}|n\rangle_A\langle n'|. \tag{2.66}$$

This operator can be obtained from the joint state by taking the *partial trace* over B, i.e. by taking the sum

$$\hat{\rho}_A = \sum_m {}_B\langle m| \Big(|\psi\rangle_{AB}\langle\psi| \Big) |m\rangle_B = \mathrm{Tr}_B \Big[|\psi\rangle_{AB}\langle\psi| \Big]. \tag{2.67}$$

Note that $\hat{\rho}_A$ is independent of the choice of bases for A and B (just like the full trace is basis independent).

Since (2.65) holds for *any* operator on A, it follows that all the information which can be learned by observing subsystem A alone is encoded in $\hat{\rho}_A$. We call $\hat{\rho}_A$ the *reduced state* of system A. Similarly, we can define a reduced state $\hat{\rho}_B = \mathrm{Tr}_A[|\psi\rangle_{AB}\langle\psi|]$ for system B, which encodes all that can be learned given acces to B alone. In general, the information contained in $\hat{\rho}_A$ and $\hat{\rho}_B$ together is not equivalent to that of the joint state $|\psi\rangle_{AB}$, because information about the correlations between A and B is lost if only one or the other subsystem is accessible. Several different joint states $|\psi\rangle_{AB}$ may be compatible with the same pair of reduced states $\hat{\rho}_A$, $\hat{\rho}_B$.

Example 2.3: Reduced states of entangled qubits

Consider two entangled two-qubit states

$$|\phi\rangle = \frac{1}{\sqrt{2}}(|0,0\rangle_{AB} + |1,1\rangle_{AB}), \tag{2.68}$$

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0,1\rangle_{AB} + |1,0\rangle_{AB}). \tag{2.69}$$

These states are not the same, in fact they are orthogonal $\langle\phi|\psi\rangle = 0$. They also encode different correlations between the subsystems. If both qubits are measured in the computational basis, for $|\phi\rangle$ the outcomes on A and B will also be identical while for $|\psi\rangle$ they will always be opposite. However, it is easy to see that the reduced states are the same

$$\hat{\rho}_A = \mathrm{Tr}_B|\phi\rangle\langle\phi| = \mathrm{Tr}_B|\psi\rangle\langle\psi| = \frac{1}{2}\mathbb{1}_A, \tag{2.70}$$

and similarly for B.

Note that the reduced state is *not* a ray in Hilbert space and thus not a pure state. Rather, it is a density operator.

2.3.4 Density operators

The reduced state (2.66) has unit trace $\mathrm{Tr}\hat{\rho}_A = \sum_{n,m} |\alpha_{nm}|^2 = 1$ (assuming that the joint state is normalised), it is hermitian $\hat{\rho}_A^\dagger = \hat{\rho}_A$, and it is positive semi-definite (the proof is left as Prob. 2.11). In fact, any operator with these properties can be understood as the reduced state of some larger system.

A *density operator* (or *density matrix*) is any operator $\hat{\rho}$ on a complex Hilbert space with the properties

$$\text{(i)} \quad \mathrm{Tr}[\hat{\rho}] = 1 \qquad \text{(unit trace).} \tag{2.71}$$

$$\text{(ii)} \quad \hat{\rho} \geq 0 \qquad \text{(positive semidefiniteness).} \tag{2.72}$$